

TUNKU ABDUL RAHMAN UNIVERSITY OF MANAGEMENT AND TECHNOLOGY

FACULTY OF COMPUTING AND INFORMATION TECHNOLOGY

ACADEMIC YEAR 2025/2026

OCTOBER EXAMINATION

**BMIT3084 ENTERPRISE NETWORKING**

TUESDAY, 14 OCTOBER 2025

TIME: 2.00 PM – 4.00 PM (2 HOURS)

BACHELOR OF INFORMATION TECHNOLOGY (HONOURS) IN INFORMATION SECURITY  
BACHELOR OF INFORMATION TECHNOLOGY (HONOURS) IN INTERNET TECHNOLOGY  
BACHELOR OF INFORMATION TECHNOLOGY (HONOURS) IN SOFTWARE SYSTEMS  
DEVELOPMENT

**Instructions to Candidates:**

Answer **ALL** questions. All questions carry equal marks.

**BMIT3084 ENTERPRISE NETWORKING****Question 1**

You have been assigned the role of a network administrator to analyse, illustrate, and configure different types of static routes for the network topology shown in Figure 1-1, which has Internet Protocol version 4 (IPv4) addressing configured on all its devices.

All devices are addressed as shown in Figure 1-1 and basic Layer 2 connectivity is operational. ORION forwards traffic for non-local destinations to AURORA over the Ethernet link. The Internet Service Provider (ISP) routes traffic towards the ORION\_LAB and NEBULA\_LAB subnets. With these assumptions and by referring to Figure 1-1, complete the configurations requested on AURORA.

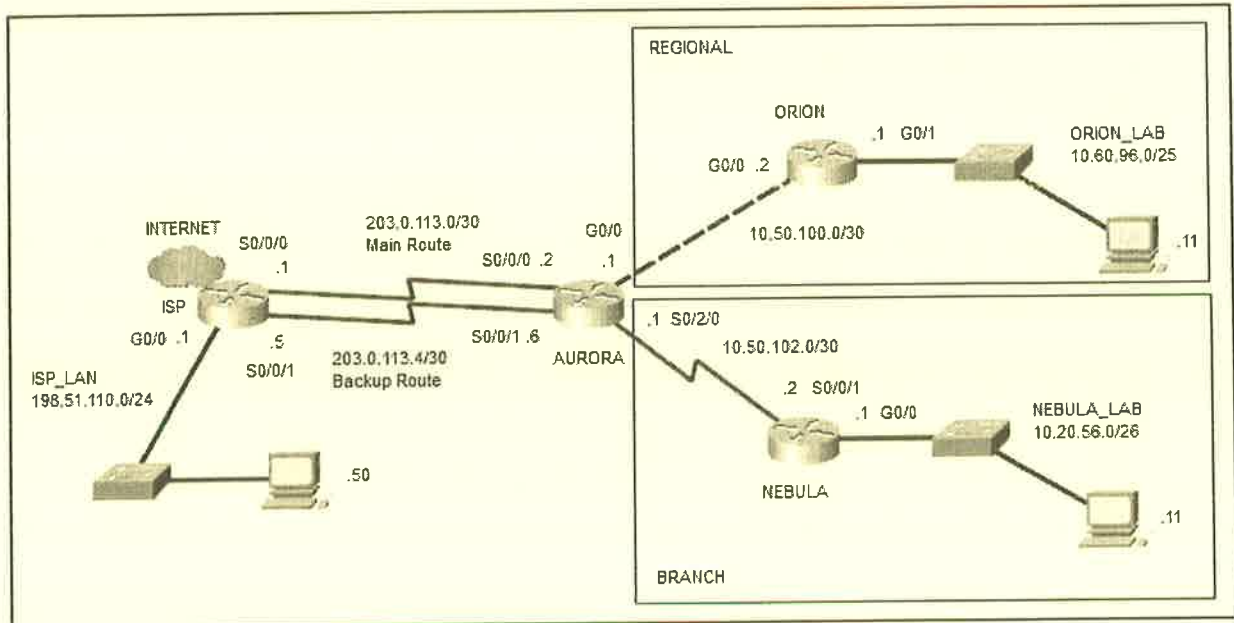


Figure 1-1: A Network Topology

- a) On **AURORA**, complete the following configurations.
- Configure a **default static route** and a **floating default static route** using **next-hop IP address** to forward traffic to the **ISP**. State your assumptions for the **floating default static route**. (4 marks)
  - Configure a **standard static route** on **AURORA** using **exit interface** to reach **NEBULA\_LAB** via **NEBULA**. (2 marks)
  - Configure a **fully specified standard static route** on **AURORA** to reach **ORION\_LAB** via **ORION**. (4 marks)
  - Explain why a **fully specified static route** is required in Question 1 a) (iii). (6 marks)

**BMIT3084 ENTERPRISE NETWORKING****Question 1 (Continued)**

- b) A network topology in Figure 1-2 was pre-configured with a **default static route** in **CORE** to forward traffic to the **ISP** and the **Internet**. The **ISP** is already using static routing. Configure **Open Shortest Path First (OSPF)** on **CORE** and **RIDGE** only as **VALLEY** is already OSPF-enabled in **area 0**. Use the network command with wildcard masks based on the subnet masks. Propagate the default route in **CORE** to **RIDGE** and **VALLEY** so that these routers can forward traffic to the **ISP** and the **Internet**. Use OSPF **process-id 88** and **area-id 0**. Use Table 1-1 to document your answer.

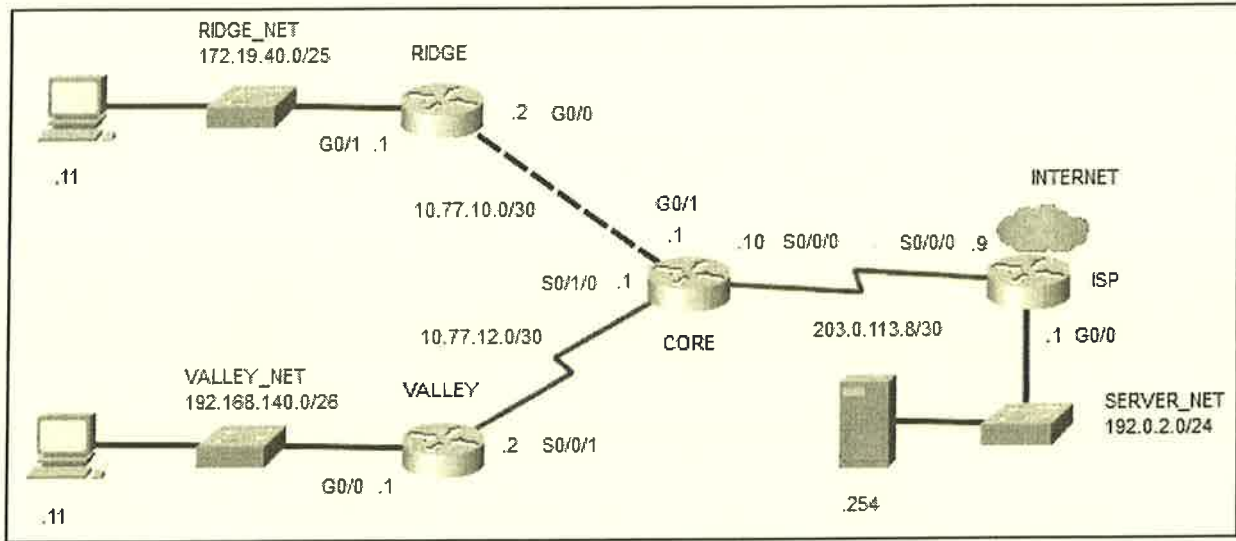


Figure 1-2: A Network Topology

Table 1-1: Documentation Table

| Router name | Configurations |
|-------------|----------------|
|             |                |

(9 marks)

[Total: 25 marks]

**Question 2**

- a) Networks face many malicious threats. Categorising these attacks by their objectives and methods helps in effective defence.
- Distinguish between Reconnaissance Attacks and Access Attacks, describing their primary objectives and at least **TWO (2)** techniques for each. (6 marks)
  - Propose **TWO (2)** methods to mitigate general network attacks within an enterprise network, referring to the principles of CIA (Confidentiality, Integrity, and Availability). (5 marks)

**BMIT3084 ENTERPRISE NETWORKING****Question 2 (Continued)**

- b) Access Control Lists (ACLs) regulate network traffic. With reference to Figure 2-1, complete the questions below. Assume basic end-to-end IP connectivity between all devices.

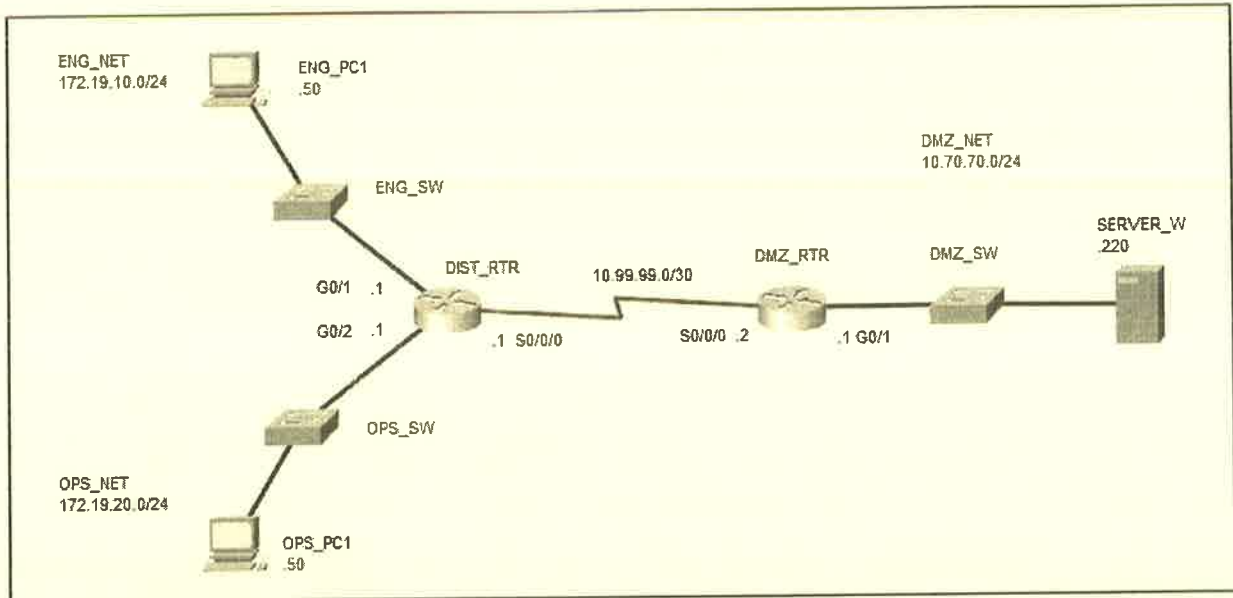


Figure 2-1: A Network Topology

Implement an IPv4 extended ACL using port number for **HTTPS** service. Write a named ACL **NET\_POLICY** that allows **HTTPS** from **OPS\_NET** to **SERVER\_W**, allows **ENG\_NET** to ping **SERVER\_W**, and explicitly denies all other traffic. Indicate the **router**, **interface**, and **direction** where the ACL should be applied. (14 marks)

[Total: 25 marks]

**BMIT3084 ENTERPRISE NETWORKING****Question 3**

The partial Dynamic Host Configuration Protocol (DHCP) configuration in Figure 3-2 was intended to provide dynamic addressing for both LANs in Figure 3-1 network topology but hosts report failures or incorrect settings.

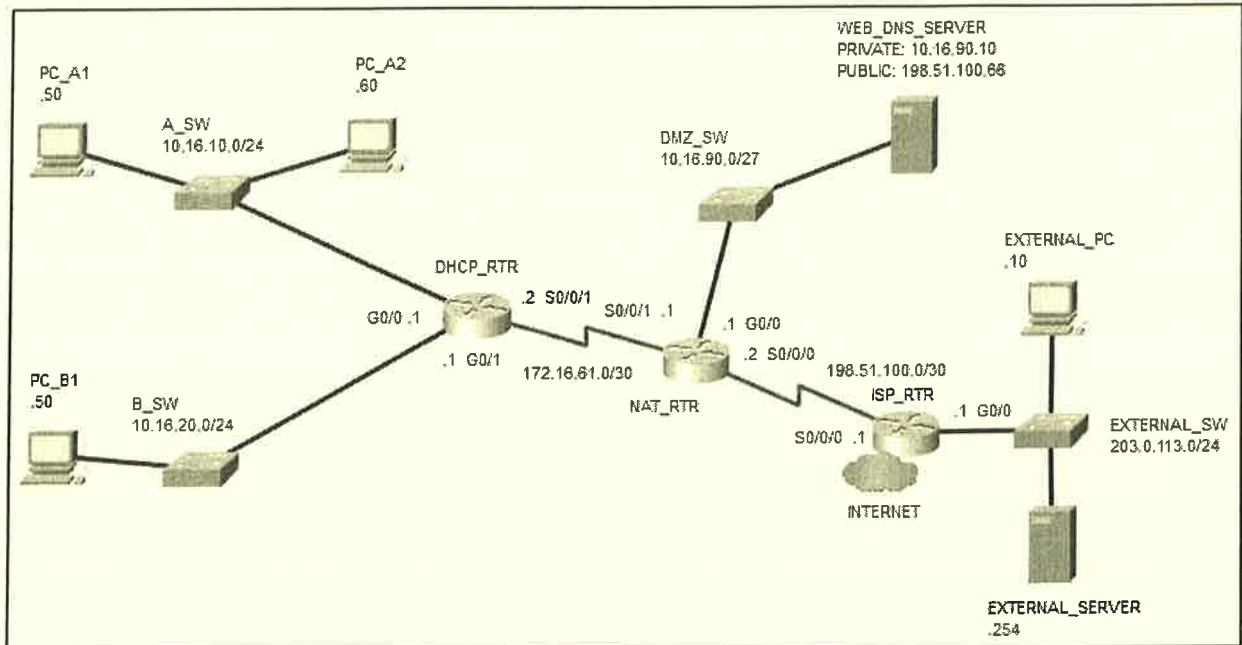


Figure 3-1: A Network Topology

**DHCP\_RTR**

```
ip dhcp excluded-address 10.16.10.100 10.16.10.200
ip dhcp excluded-address 10.16.20.100 10.16.20.200
```

```
ip dhcp pool LAN_A
network 10.16.10.0 255.255.0.0
default-router 10.16.10.254
dns-server 10.16.20.1
domain-name corp.local
```

```
ip dhcp pool LAN_B
network 10.16.30.0 255.255.255.0
default-router 10.16.20.1
dns-server 10.16.90.10
lease 3 12 0
```

Figure 3-2: Partial "show run" snippets

- a) Using Table 3-1, analyse **FOUR (4)** configuration errors in Figure 3-2 on **DHCP\_RTR**. For each error, write the correction and a brief justification. (10 marks)

Table 3-1: Documentation Table

| Errors | Solutions | Justifications |
|--------|-----------|----------------|
|        |           |                |

**BMIT3084 ENTERPRISE NETWORKING****Question 3 (Continued)**

- b) On **NAT\_RTR**, configure static NAT (Network Address Translation) so that **WEB\_DNS\_SERVER 10.16.90.10** is reachable on the public IP **198.51.100.66**. Include any necessary interface role commands for NAT to function. (5 marks)
- c) On **NAT\_RTR**, PAT (Port Address Translation) should allow **LAN\_A** and **LAN\_B** to reach the Internet via **S0/0/0** while excluding the **DMZ** from PAT. Using Table 3-2, analyse the configurations errors in Figure 3-3 and provide the corrected command(s). (10 marks)

| <b>NAT_RTR</b>                                                                                                                                                                                                                                                                                     |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <pre>access-list 100 permit ip 10.16.10.0 0.0.0.255 any access-list 100 permit ip 10.16.20.0 0.0.0.255 any  ip nat inside source list 100 interface GigabitEthernet0/0  interface Serial0/0/1 ip nat outside interface Serial0/0/0 ip nat inside interface GigabitEthernet0/0 ip nat outside</pre> |

Figure 3-3: Router NAT\_RTR (NAT/PAT excerpt)

Table 3-2: Documentation Table

| <b>Errors</b> | <b>Solutions</b> |
|---------------|------------------|
|               |                  |

[Total: 25 marks]

**Question 4**

- a) Enterprise networks employ various WAN topologies.
- Differentiate the Hub-and-Spoke Topology from the Dual-homed Topology for Wide Area Networks (WANs), explaining their key characteristics and illustrating with a simple diagram for each. (6 marks)
  - Suggest **ONE (1)** key disadvantage for each of the Hub-and-Spoke and Dual-homed WAN topologies. (4 marks)
- b) Identify the **TWO (2)** main enterprise VPN implementations and describe **ONE (1)** distinguishing characteristic of **each** implementation. (6 marks)

**BMIT3084 ENTERPRISE NETWORKING****Question 4 (Continued)**

- c) (i) Analyse the roles of Classification and marking tools and Congestion management tools in a Quality of Service (QoS) implementation. (4 marks)
- (ii) In your analysis, differentiate between First-In, First-Out (FIFO) and Class-Based Weighted Fair Queuing (CBWFQ) as congestion management techniques. (5 marks)

[Total: 25 marks]